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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Department of Computer Science and Engineering

Course Code & Name	CSA06 – Design and Analysis of Algorithms
Assessment	CO1 – Assessment Tool 1: IT Application-Based Problem Solving – Scalability Audit of an E-Commerce Platform
Course Outcome Assessed	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method.
Duration	3 Days
Mode	Individual
Student Reg. No.	192565011
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1.What is this assignment about?

Imagine an online shopping website called ShopSphere, similar to Amazon or Flipkart.
Currently, the website has 100,000 products. Within one year, it expects to have 1,000,000 products.
The company wants to know:

- Which algorithms work quickly?
- Which algorithms become slow when the number of products increases?
- How many operations does each algorithm perform?
- How long will each algorithm take to run?
- Which algorithms should be redesigned?

Your job is to act as an algorithm engineer and solve these problems mathematically.

Introduction

The assignment describes an online shopping platform called ShopSphere.
Currently, ShopSphere has 100,000 products. Within one year, the number of products is expected to increase to 1,000,000.
As the number of products increases, some algorithms may take more time to complete their work.
The company wants to check whether its existing algorithms can handle this growth efficiently.
Your role is to analyse the algorithms, calculate their running time mathematically and recommend improvements.
The assignment assumes that the server can execute 10^8 basic operations per second. You must use this value to estimate the execution time of the algorithms.

2. Important concepts you need to understand

Algorithm

An algorithm is a step-by-step procedure used to solve a problem.
For example, searching for a product in an online shopping website is an algorithmic task.

Input size

Input size is the amount of data processed by an algorithm. It is represented by \(n\).
For example:

- If a catalogue contains 100 products, (n=100).
- If it contains 1,000 products, (n=1000).
- If it contains 1,000,000 products, (n=1,000,000).

Time complexity

Time complexity describes how the number of operations increases when the input size increases.
The main complexities in this assignment are:

- O(1): Constant time.
- O(log n): Logarithmic time.
- O(n): Linear time.
- O(n log n): Linearithmic time.
- O(n^2): Quadratic time.

For example, an \(O(n)\) algorithm generally grows more slowly than an \(O(n^2)\) algorithm when the input becomes very large.

Space complexity

Space complexity describes how much additional memory an algorithm needs while running.

Asymptotic notation

Asymptotic notation describes the growth rate of an algorithm for large input sizes.

- Big O, (O): Upper bound on growth.
- Big Omega, (Omega): Lower bound on growth.
- Big Theta, (Theta): Tight bound on growth.

These concepts are important because the assignment asks you to compare algorithms and determine which ones are efficient.

2. Understand the five modules

The assignment contains five main modules. Each performs a different job on the shopping website.

M1 – Duplicate-Listing Detector

Finds duplicate products uploaded by sellers.
Main concept: Nested loops and comparison counting.

M2 – API Rate Limiter

Checks incoming API requests in increasingly large groups.
Main concept: Loops, geometric series and asymptotic complexity.

M3 – Product Search

Compares two ways of finding a product: sequential search and binary search.
Main concept: Best, average and worst cases.

M4 – Recommendation Engine
Processes customer and product data to recommend products. The current design duplicates work across four partitions.
Main concept: Recurrence relations and Master Theorem.

M5 – Sales Analytics Engine
Recursively processes and sorts sales records by revenue.
Main concept: Recurrence relations and recursion trees.

3. Explanation of each question (a) to (g)

(a) Understand the problem
You must identify three things for each module:

1. **Input size**: How many products or records are processed? Usually represented by n.
2. **Basic operation**: The main action repeated by the algorithm.
3. **Growth rate**: How the number of operations changes as n increases.

For example, M1 compares product titles and seller names. Since it compares pairs of products, its number of comparisons grows approximately as n^2.

(b) Analyse the iterative modules
This question focuses on M1 and M2.

M1: Duplicate detection
This module identifies duplicate products in the shopping catalogue.
Sometimes, a seller uploads the same product more than once. The algorithm compares each product with every product that comes after it. It checks whether both the product title and seller are the same.
The algorithm uses two nested loops. The outer loop selects a product, and the inner loop compares it with the remaining products.
As the number of products increases, the number of comparisons increases rapidly.
The number of comparisons is:

$$T(n) = (n-1) + (n-2) + \cdots + 2 + 1$$

Using the sum formula:

$$T(n) = \frac{n(n-1)}{2}$$

Therefore,

$$\boxed{T(n) = \Theta(n^2)}$$

Meaning: if the product count becomes 10 times larger, the comparisons become approximately 100 times larger.

M2: API rate limiter
This module checks requests sent to the platform's public API.
The algorithm starts with i=1. After each iteration, it doubles the value of i, producing values such as 1, 2, 4, 8, 16 and so on.
For each value of i, the inner loop performs i audit operations.
The total number of operations for a power-of-two input is:

$$T(n) = 1 + 2 + 4 + \cdots + \frac{n}{2}$$

Using the geometric series formula:

$$T(n) = n - 1$$

Therefore:

$$T(n) = \Theta(n)$$

The team member claims that the complexity is $\Theta(n \log n)$.This claim is incorrect because the inner loop does not run \(n\) times during every outer-loop iteration. Its size doubles, and the sum of those sizes is linear.

For n=1024:

$$T(1024) = 1024 - 1 = 1023$$

Thus, the algorithm performs 1,023 audit operations.

(c) Solve a recurrence by substitution

This question asks you to solve search version S2, which uses binary search.
Binary search repeatedly divides the search area into half.
The given recurrence is:

$$T(n) = T(n/2) + 1, \quad T(1) = 1$$

Using backward substitution:

$$T(n) = T(n/4) + 2$$
$$T(n) = T(n/8) + 3$$

After k steps:

$$\frac{n}{2^k} = 1$$
$$T(n) = T(n/2^k) + k$$

Stop when the remaining input size is 1:

$$\frac{n}{2^k} = 1$$

Therefore,

$$k = \log_2 n$$

Hence,

$$\boxed{T(n) = 1 + \log_2 n = \Theta(\log n)}$$

The question also asks you to compare the space complexity of iterative and recursive binary search. Iterative binary search uses O(1) auxiliary space, while recursive binary search uses \(O(\log n)\) stack space.

(d) Apply the Master Theorem

This section deals with M4 and M5.
The Master Theorem helps you find the complexity of recursive algorithms without expanding every step manually.
The general formula is:

$$T(n) = aT(n/b) + f(n)$$

Here:

- a = number of recursive subproblems
- b = factor by which the input size is divided
- f(n) = work done outside recursive calls

M4: Current recommendation engine

This module processes customer and product data to recommend products to customers.
The current design divides the problem into four recursive subproblems, each of size \(n/2\). It then combines the results in linear time.
The recurrence relation is:

$$T(n) = 4T(n/2) + n$$

Calculate:

$$n^{\log_b a} = n^{\log_2 4} = n^2$$

The recursive work grows faster than the additional linear work \(n\). Applying Master Theorem Case 1 gives:

$$\boxed{T(n) = \Theta(n^2)}$$

Proposed redesign:

$$T(n) = 2T(n/2) + n$$

Now,

$$n^{\log_2 2} = n$$

This matches f(n)=n, so Master Theorem Case 2 gives:

$$\boxed{T(n) = \Theta(n \log n)}$$

The proposed design is asymptotically more efficient.

M5: Sales analytics

This module processes sales data and sorts records according to revenue.
The algorithm divides the data into two halves, processes each half recursively and performs additional sorting work at each level.
The recurrence relation is:

$$T(n) = 2T(n/2) + n \log n$$

This recurrence requires the extended Master Theorem or a recursion-tree analysis. The resulting complexity is:

$$\boxed{T(n) = \Theta(n \log^2 n)}$$

(e) Analyse best, average and worst cases

You must compare S1 and S2.

Case	Meaning
Best case	The algorithm finds the product with the fewest comparisons.
Average case	The expected number of comparisons over the possible search outcomes.
Worst case	The algorithm uses the maximum number of comparisons.

S1 – Sequential search

- Best case: Product is first, requiring 1 comparison.
- Worst case: Product is last or absent, requiring up to \(n\) comparisons.
- Average case: Depends on the probability \(p\) that the product exists.

If the product exists with probability \(p\), and its position is uniformly distributed, then:

$$E[C] = p\frac{n+1}{2} + (1-p)n$$

For \(n=1,000,000\) and \(p=0.5\):

E[C]=0.5(500000.5)+0.5(1000000)

E[C]=750000.25

S2 – Binary search

- Best case: The product is at the middle entry, requiring 1 comparison.
- Average case: Typically$\Theta(\log n)$.
- Worst case: $\Theta(\log n)$assuming the index is sorted and the input size is suitable.

Binary search is generally much faster for large catalogues, but it requires sorted data.

(f) Prove an asymptotic bound

The report generator has the function:

$$f(n) = 3n^2 + 10n\log_2 n + 50$$

You must prove that:

$$\boxed{f(n) = \Theta(n^2)}$$

The formal definition requires positive constants c1,c2 and a threshold $n_0$such that:

$$c_1 n^2 \leq f(n) \leq c_2 n^2$$

for every $n \geq n_0$

One valid choice is \($c_1 = 3$ $c_2 = 63$, and $n_0 = 2$For $n \geq 2$,$\log_2 n \leq n$and the expression can be bounded above by \(63n^2\).

You must also show that f(n) is not O(n). This is because the quadratic term 3n^2 grows faster than any constant multiple of n.

(g) Scalability decision

This is the final practical decision-making question.

You must calculate the number of operations and estimated running time for M1, M2, S2, both versions of M4, and M5 when n=1,000,000.

The estimated time is:

$$\text{Time} = \frac{\text{Number of operations}}{10^8}$$

seconds.

For example, M1 has:

$$T(n) = \frac{n(n-1)}{2}$$

At one million products:

$$T(n) = 499{,}999{,}500{,}000$$

Estimated time:

$$\frac{499{,}999{,}500{,}000}{10^8} = 4999.995 \text{ seconds}$$

That is approximately 83 minutes 20 seconds, which exceeds the 10-minute nightly processing window.

These are estimates based on the stated operation-count model. Actual runtime also depends on implementation, hardware, memory access and other overheads.

4. The most important formulas to remember

Concept	**Formula**	**Complexity**
Nested pair comparisons	$\frac{n(n-1)}{2}$	$\Theta(n^2)$
Doubling loop	$1 + 2 + \cdots + n/2$	$\Theta(n)$
Binary search	$\log_2 n$	$\Theta(logn)$

Redesigned M4	2T(n/2)+n	$\Theta(nlogn)$
Current M4	4T(n/2)+n	$\Theta(n^2)$
M5	2T(n/2)+nlog n	$\Theta\left(\Theta(n\log^2 n)\right)$

5. How to complete the assignment

1. Start with section (a), identifying the input size and basic operation.
2. Show the full mathematical derivations in sections (b), (c) and (d).
3. Explain best, average and worst cases in section (e).
4. Write the formal proof in section (f).
5. Finish with the operation counts, time estimates and redesign recommendations in section (g).

Mark Evaluation:

Criteria	Weightage (%)	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20				
Application of Concepts	30				
Problem-Solving Approach	20				
Accuracy of Solution	20				
Clarity	10				
Total Marks					

Code of Conduct

I _____ (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ Date: _____